Ideation Phase Brainstorm & Idea Prioritization Template

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Team ID	SWUID20264931
Project Name	Electric vehicle Charge and Range
Maximum Marks	

Brainstorm & Idea Prioritization

Topic: EV Charge and Range

1. Problem Statement

Electric Vehicles (EVs) are becoming popular as a sustainable transportation option. However, users often face challenges related to **charging time, battery range, and charging infrastructure availability**. These issues create **range anxiety**, making consumers hesitant to adopt EVs.

The goal of this brainstorming session is to generate ideas that can **improve EV charging efficiency and increase driving range** while enhancing user convenience.

2. Brainstorming Ideas

Idea No.	Idea Title	Description
1	Fast Charging Technology	Develop ultra-fast charging stations that can charge EV batteries within 10–15 minutes.
2	Smart Battery Management System	Use AI-based battery monitoring to optimize battery performance and extend driving range.
3	Solar Charging Stations	Install solar-powered charging stations to reduce electricity cost and provide sustainable charging.
4	Mobile Charging Vans	Introduce mobile EV charging services that can assist vehicles running out of charge on the road.
5	Battery Swapping Stations	Create stations where users can swap their discharged battery with a fully charged one instantly.
6	Range Prediction App	Develop an app that predicts remaining driving range based on driving style, weather, and traffic.

Idea No.	Idea Title	Description
7	Wireless Charging Roads	Implement wireless charging infrastructure in roads where vehicles charge while driving.
8	Charging Station Locator System	Build a smart navigation system that shows nearby charging stations and their availability.

3. Idea Evaluation Criteria

To prioritize the ideas, we evaluate them based on the following factors:

- **Feasibility** – How easy it is to implement
- **Cost** – Implementation and operational cost
- **Impact** – How much it improves EV charging and range
- **User Convenience** – Ease of use for EV drivers

4. Idea Prioritization Table

Idea	Feasibility	Cost	Impact	Priority
Fast Charging Technology	Medium	High	High	High
Smart Battery Management System	High	Medium	High	Very High
Solar Charging Stations	Medium	Medium	Medium	Medium
Mobile Charging Vans	High	Medium	Medium	High
Battery Swapping Stations	Medium	High	High	High
Range Prediction App	High	Low	Medium	Very High
Wireless Charging Roads	Low	Very High	High	Low
Charging Station Locator	High	Low	High	Very High

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5. Top Priority Ideas

Based on evaluation, the most promising solutions are:

1. **Smart Battery Management System** – Improves battery life and driving range.
2. **Range Prediction Application** – Helps drivers plan trips effectively.
3. **Charging Station Locator System** – Reduces range anxiety.
4. **Fast Charging Infrastructure** – Reduces charging time significantly.

6. Expected Benefits

- Increased EV adoption
- Reduced range anxiety among drivers
- Improved energy efficiency
- Better charging infrastructure management
- Enhanced user experience